BUPIVACAINE

Cardiotoxic
Never IV

CHEMICAL NAME 1-butyl-N-(2,6-dimethylphenyl)piperidine-2-carboxamide (as hydrochloride)
CLASS Long-acting amide local anaesthetic
TRADE NAMES Marcaine · Sensorcaine · Marcain Heavy (spinal) · Exparel (liposomal)

THERAPEUTIC USES / INDICATIONS

- Spinal (subarachnoid) anaesthesia — heavy (hyperbaric) 0.5% for caesarean section and lower-limb surgery.
- Epidural analgesia in labour and post-operatively.
- Peripheral nerve blocks and wound infiltration for long-lasting analgesia.

ADMINISTRATION

- Given by anaesthetist/trained clinician only.
- Never give IV; always aspirate before injection.
- Epidural infusion: label lines clearly as epidural (yellow), use dedicated pumps.
- 0.75% contraindicated in obstetric epidural anaesthesia.

DOSAGES

- **Max dose:** 150 mg single dose; no more than 2 mg/kg in any 4 h.
- **Spinal (heavy 0.5%):** 2–4 mL (10–20 mg).
- **Labour epidural:** low-concentration bupivacaine (often with fentanyl) per obstetric anaesthesia protocol.
- **Infiltration:** 0.25%.

PHARMACOKINETICS

- **Onset:** 5–10 min (spinal); 10–20 min (epidural). **Duration:** 3–8 h (longest of common agents).
- Highly protein-bound; metabolised in liver.
- Excreted in urine. **t½:** 2.7 h (adults), up to 8 h in neonates.

PHARMACODYNAMICS

- Blocks **sodium channels** → prevents nerve conduction.
- Lower concentrations give sensory block with less motor block (useful in labour).
- Binds cardiac sodium channels strongly → cardiotoxicity difficult to reverse.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hypotension (spinal/epidural sympathetic block).
- Nausea, vomiting; shivering.
- Urinary retention; motor weakness of legs.

ADVERSE EFFECTS (SERIOUS)

- Cardiotoxicity: ventricular arrhythmias, cardiac arrest (treat with lipid emulsion 20%).
- Seizures (systemic toxicity).
- High or total spinal block → respiratory arrest.
- Post-dural puncture headache; epidural haematoma or abscess.

CONTRAINDICATIONS & CAUTIONS

- IV regional anaesthesia (Bier's block).
- Allergy to amide anaesthetics.
- Spinal/epidural: coagulopathy, anticoagulation, infection at site, severe hypovolaemia, raised ICP.
- 0.75% in obstetrics.

KEY INTERACTIONS

- **Anticoagulants (LMWH, DOACs):** timing rules for epidural insertion/removal.
- **Antiarrhythmics:** additive cardiac toxicity.

NURSING MANAGEMENT

- After spinal/epidural: monitor BP every 5 min initially, HR, SpO_2, sensory level and motor block.
- Have IV fluids, vasopressor (ephedrine/phenylephrine) and lipid emulsion available.
- Check epidural site; monitor bladder (catheter may be needed).
- Assess legs for return of movement; report persistent weakness or back pain (haematoma).
- Never connect IV medications to epidural line.

PATIENT EDUCATION

- Legs will feel heavy and numb for a few hours — ask for help before standing.
- Report headache when sitting up, back pain or new numbness.
- Tell the nurse about any ringing in the ears or metallic taste.

KEY POINTS

- Long-acting local anaesthetic for spinal/epidural.
- Max 2 mg/kg single dose.
- Most cardiotoxic; never IV.
- Monitor BP, sensory/motor block after neuraxial use.

SOURCES CHECKED [77] Bupivacaine Heavy 5 mg/ml Solution for Injection (UK SmPC) [78] Bupivacaine 5mg/ml Solution for Injection (UK SmPC) *Full references p. 432–457*